

PROJECT MANAGEMENT

Principles of Project Management: Agile, Scrum & Traditional

From Scope Planning, WBS & Critical Path Method to Agile Sprints, Risk Management & Leadership

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CHAPTER 1

Project Initiation, Scope Definition & Work Breakdown (WBS)

1. Theoretical Foundations & Scope

Successful project management begins with clear objective definition and stakeholder alignment. This chapter covers project charters, SMART goals, requirements gathering, scope creep prevention, and Work Breakdown Structures.

2. Core Conceptual Framework

- **Project Charter & Stakeholders:** Authorizing the project, identifying key influencers, and establishing success metrics.
- **Work Breakdown Structure (WBS):** Hierarchical decomposition of total project scope into manageable, assignable work packages.
- **Scope Baseline & Change Control:** Formal processes for evaluating, approving, and documenting changes to project deliverables.

Key Practical Insights & Examination Takeaways:

- ' Draft actionable project charters with clearly articulated business cases.
- ' Construct 100% rule compliant Work Breakdown Structures.
- ' Implement change control boards to prevent unmanaged scope creep.

CHAPTER 2

Project Scheduling: Gantt Charts, CPM & PERT Analysis

1. Theoretical Foundations & Scope

Timely project delivery requires rigorous dependency mapping and scheduling. We examine activity sequencing, Precedence Diagramming Method (PDM), Critical Path Method (CPM), forward/backward pass calculations, and float analysis.

2. Core Conceptual Framework

- **Critical Path Method (CPM):** Determining the sequence of dependent activities that dictates the shortest possible project duration.
- **Total Float vs Free Float:** Calculating the amount of time an activity can slip without delaying the successor or final deadline.
- **PERT Three-Point Estimation:** Weighted averaging of optimistic, most likely, and pessimistic task durations under uncertainty.

Key Practical Insights & Examination Takeaways:

- ' Identify critical path tasks where delays directly jeopardize overall completion.
- ' Create Gantt charts showing concurrent and sequential project milestones.
- ' Perform resource leveling to resolve over-allocated project team members.

CHAPTER 3

Agile Methodologies, Scrum Framework & Risk Management

1. Theoretical Foundations & Scope

Modern software and technology projects rely heavily on iterative Agile frameworks. This chapter covers the Agile Manifesto, Scrum ceremonies (Sprint Planning, Daily Standup, Sprint Review, Retrospective), and proactive risk registers.

2. Core Conceptual Framework

- **Scrum Roles & Artifacts:** Product Owner, Scrum Master, and Developers collaborating around Product Backlog, Sprint Backlog, and Increment.
- **Sprint Velocity & Burndown Charts:** Tracking team velocity using story points and identifying delivery bottlenecks.
- **Risk Probability & Impact Matrix:** Categorizing project risks, establishing mitigation triggers, and maintaining contingency reserves.

Key Practical Insights & Examination Takeaways:

- ' Facilitate effective Scrum events and write clear User Stories with acceptance criteria.
- ' Assess qualitative and quantitative risks using probability-impact scoring.
- ' Cultivate psychological safety and servant leadership in cross-functional teams.

Academic References & Further Study

[1] Quality Education Academic Standards & Pedagogical Guidelines, 2026 Edition.

[2] International Computer Science & Engineering Curriculum Framework.

[3] Educational Technology & Assessment Methodology Research Series.

[4] Open Access Educational Resource Repository - Quality Education Platform.

